

COMPLEXITY ANALYSIS

Time Complexity

Time complexity tells us how fast the program runs as the input size grows. I analyzed each part of my program.

Input and Initial Processing

Input: For n students, I need to get roll, name, and 5 marks. This takes $O(n)$ time. Each student takes constant time.

calculate() function: Goes through each student and adds 5 marks. This also takes $O(n)$ time.

Total: $O(n) + O(n) = O(n)$

Merge Sort

Merge Sort always takes $O(n \log n)$ time, regardless of the input.

For $n = 100$:

- $n \log n = 100 \times 6.64 = 664$ operations
- This is very fast

Binary Search

Binary Search takes $O(\log n)$ time.

For $n = 100$:

- $\log n \approx 6.64$ comparisons
- Very fast

Leaderboard

The leaderboard uses bubble sort on the copy. Bubble sort is $O(n^2)$.

For $n = 100$:

- $n^2 = 10,000$ operations
- Still fast for 100 students

But if n becomes very large (like 10,000), bubble sort would be slow. For my use case (up to 100 students), it's fine.

Overall Time Complexity

The most expensive operation is Merge Sort ($O(n \log n)$) or Leaderboard ($O(n^2)$).

For $n \leq 100$, both are fast enough.

Space Complexity

Space complexity tells us how much memory the program uses.

Arrays

- s array: $n \times \text{size of Student}$
- temp array: $n \times \text{size of Student}$
- For $n = 100$: about 12 KB

Leaderboard Copy

The leaderboard function creates another copy:

- top array: $n \times \text{size of Student}$
- For $n = 100$: about 6 KB

Total Space

Total space needed: about 18 KB maximum.

This is very small. Modern computers have GBs of RAM.

Complexity Summary Table

Operation	Time Complexity	Space Complexity
Input	$O(n)$	$O(1)$
calculate()	$O(n)$	$O(1)$
Merge Sort	$O(n \log n)$	$O(n)$
Binary Search	$O(\log n)$	$O(1)$
Leaderboard	$O(n^2)$	$O(n)$
Display	$O(n)$	$O(1)$

What This Means

For small to medium class sizes (up to 100 students), the program will be very fast. All operations will complete in a fraction of a second.

The slowest part is the leaderboard if we had a large number of students. But for typical school use, this is not a problem.